

Carbon dioxide sequestration on biocement-based composites

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Mondem S. Reddy, Sumit Joshi

Thapar Institute of Engineering & Technology, Patiala, India

10.1 Introduction

Rapid industrial expansion around the world to achieve stable economic growth has led to adverse impact on the global climate system. A large quantity of green house gases (GHG) has accumulated in the earth's atmosphere due to anthropogenic emission. Increased concentrations of GHG in earth's atmosphere mainly consist of carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), and chlorofluorocarbons ([National Research Council's, 2001](#)). The increase in GHG emissions in the global atmosphere led to increased average global temperature. Among the radiative GHG, CO_2 contributes maximum radiative forcing and affects the earth-atmosphere energy balance ([Halmann and Steinberg, 1999](#)). According to Intergovernmental Panel on Climate Change (IPCC), a cumulative CO_2 emission since 1970 has tripled because of fossil fuel combustion and cement industry ([IPCC, 2014](#)). A human-induced CO_2 emission has severely damaged the climate system, causing irreversible impacts such as thermal expansion of oceans due to warming ([Solomon et al., 2008](#)). According to IPCC, the period from 1983 to 2012 in the northern hemisphere was the warmest 30-year period of the last 800 years ([IPCC, 2014](#)). Global mean sea level rise of about 75% was reported due to glacier mass loss and ocean thermal expansion from warming. Global warming has led to sensitive change in average global precipitation, which will adversely affect ecosystems, agriculture, and human health ([Zhang et al., 2007](#)). However, both land and ocean are the main contributors in the removal of atmospheric CO_2 and act as natural sink. With the increased level of CO_2 concentration, expected decrease in the efficiency of both land and ocean sinks because of CO_2 fertilization limits on land and decrease in carbonate concentration, which buffers CO_2 in the ocean ([Le Quéré et al., 2009](#)). Oceanic uptake of atmospheric CO_2 has resulted into gradual acidification of ocean surface water, which further affects coral reefs and marine biota ([Kleypas et al., 1999](#)). Carbon sequestration in land sink by terrestrial ecosystem is also effected due to increased level of CO_2 ([Bradford, 2017](#)). In order to mitigate the adverse effects of increasing global CO_2 concentration, several approaches are initiated all around the globe.

To counter the alarming increase of CO_2 concentration, approaches such as capture, storage, and utilization of CO_2 were adopted. Different approaches for CO_2 capture and storage have been associated on the basis of combustion processes. CO_2 capture approaches are classified as post-conversion, preconversion, and oxy-fuel

combustion (Leung et al., 2014). Post-conversion is a technique in which carbon is captured from flue gas generated after the combustion of fossil fuels. Post-conversion capture method includes absorption in solvents (such as monoethanolamine, diethanolamine, zeolite, and activated carbon) for separation of CO₂ (Aaron and Tsouris, 2005). In preconversion CO₂ capture, the fossil fuel (coal or natural gas) is treated initially before combustion. Coal is treated through gasification process under low O₂ environment, resulting into the formation of syngas (CO + H₂) (Olajire, 2010). Post-conversion and preconversion techniques are applied to power plants in which fossil fuels are used as energy source. In the oxy-fuel combustion process fuel is burned in pure oxygen, resulting into high CO₂ concentrations free from NO and NO₂ compounds. This avoids the need for chemicals or other means for the separation of CO₂ from the flue gas (Azapajic et al., 2004). However, the ease and cost economics in the abovementioned carbon separation and capture approaches is a matter of concern (D'Alessandro et al., 2010).

Relocation of captured CO₂ by injection into geologic or oceanic sinks has been an option for disposal methods. Capabilities of CO₂ sequestration into sinks such as ocean, deep saline, oil and gas reservoirs, and coal seams are reported (Leung et al., 2014). At present, geological storage is considered to be the most viable option to store large quantities of CO₂. These geological formations can store collectively hundreds of gigatons of carbon (Herzog et al., 2000). Deep saline aquifers at 700–1000 m below ground level are particularly promising option for the sequestration of large quantities of CO₂. The Sleipner natural gas field in the North Sea off the coast of Norway represents the first time that 1.1 Mt of CO₂/year has been stored underground in the deep saline aquifer (White et al., 2003). CO₂ storage and natural gas recovery in unmineable coal seams were also reported. The concept of enhanced coalbed methane recovery with CO₂ was first developed in 1991 by researchers from the Alberta Research Council. A small CO₂ pilot injection was performed in the Fruitland Coal Formation in southern Colorado to recover CH₄ (White et al., 2003). Deep Ocean has also been reported as largest potential reservoir to store CO₂. Effective ocean sequestration for CO₂ was reported by injecting CO₂ into the sea below thermocline layer of ocean (Herzog et al., 2000). Technical and environmental issues such as leakage of CO₂ in geological storage are also associated. High cost of capture and storage of CO₂ in the abovementioned approaches is also a demerit.

The Paris climate change agreement marks a new era of optimism and action to cater the record levels of high greenhouse gases. Various countries have implemented carbon trading and taxation to tackle increasing carbon emissions (Princiotta, 2007). Boone et al. (2013) reviewed the progress made in developing carbon capture technologies, and IPCC has also published different carbon storage and capture technologies for reducing GHG emissions (IPCC, 2014). Some reviews have also outlined different approaches for carbon capture and also described methods for mitigating global warming (Bose and Satyanarayana, 2017). This review discusses biomimetic carbon sequestration on biocement-based composites through biomineralization process and their applications in civil engineering.

10.2 Biocement

10.2.1 *Microbes involved in biocement*

Biocementation is an ecological process based on microbial induced carbonate precipitation (MICP) mechanism, which results in the deposition of calcium carbonate. Different bacterial species in natural habitats precipitate carbonates in alkaline environments rich in Ca^{2+} ions with various mechanisms (Ehrlich, 1998). Boquet et al. (1973) reported that calcium carbonate formation under suitable conditions is a common phenomenon for almost all bacteria. At neutral pH, presence of carboxyl, phosphoryl, and amino groups at the bacterial cell surface results in heterogeneous electronegativity charge, which is later responsible for nucleating site favoring the adsorption of positively charged cations (e.g., Ca^{2+} , Mg^{2+}). Presence of calcium ions in the surroundings of bacterial cell wall results into the precipitation of calcium carbonate (Douglas and Beveridge, 1998). Four groups of microorganisms, such as photosynthetic organisms (cyanobacteria and algae), sulfate reducing bacteria (SRB) (dissimilatory reduction of sulfates), organisms utilizing organic acids, and organisms involved in the nitrogen cycle, are reported to be involved in the process of biocementation.

10.2.2 *Routes of biomineralization*

Broadly two different metabolic pathways are involved in the process of biomineralization associated with the microorganisms: (1) autotrophic pathway; (2) heterotrophic pathway.

10.2.2.1 *Autotrophic mediated pathway*

This is the conversion of CO_2 into calcium carbonate crystals by microbes in the presence of CO_2 and calcium source. Autotrophic precipitation of carbonates include oxygenic and anoxygenic photosynthesis and nonmethylootropic methanogenesis (Castanier et al., 1999). All of the three autotrophic pathways use CO_2 as a carbon source. Bacterial calcium carbonate precipitations with aforementioned autotrophic pathways with listed mechanism are shown in Table 10.1.

In nonmethylootropic methanogenesis pathway, methanogenic archaeobacteria utilizes CO_2 and H_2 to give methane in the absence of oxygen (Castanier et al., 1999). Anaerobic oxidation of methane results in the formation of bicarbonate by electron acceptor sulfate as shown in Eq. (10.2). Carbonate in the presence of calcium ions results into the precipitation of calcium carbonate as shown in Eq. (10.3). In anoxygenic photosynthesis, H_2S acts an electron donor in the formation of methanol carried out by purple and green photosynthetic bacteria as shown in Eq. (10.4) (Castanier et al., 2000). Oxygenogenic photosynthesis is carried out by cyanobacteria using visible light (680–685 nm) as a source of energy leading into the production of oxygen. In this process, water acts as an electron donor as shown in Eq. (10.7) (Ehrlich, 1998). Oxygenic photosynthetic pathway associated with cyanobacteria in marine and

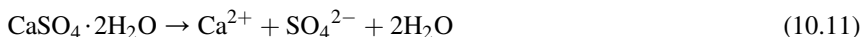
Table 10.1 Different mechanisms of calcium carbonate precipitation by autotrophic bacteria

Metabolic pathway	Mediator	Mechanism	Equation no.	References
Nonmethylotrophic methanogenesis	Methanogenic archaeobacteria	$\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$	(10.1)	Castanier et al. (1999)
		$\text{CH}_4 + \text{SO}_4^{2-} \rightarrow \text{HCO}_3^- + \text{HS}^- + \text{H}_2\text{O}$	(10.2)	
		$\text{Ca}^{2+} + 2\text{HCO}_3^- \leftrightarrow \text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O}$	(10.3)	
Anoxygenogenic photosynthesis	Sulfurous or nonsulfurous purple and green photosynthetic bacteria	$\text{CO}_2 + 2\text{H}_2\text{S} + \text{H}_2\text{O} \rightarrow \text{CH}_2\text{O} + 2\text{S} + 2\text{H}_2\text{O}$	(10.4)	Castanier et al. (2000)
		$2\text{HCO}_3^- \leftrightarrow \text{CO}_2 + \text{CO}_3^{2-} + \text{H}_2\text{O}$	(10.5)	
		$\text{CO}_3^{2-} + \text{H}_2\text{O} \leftrightarrow \text{HCO}_3^- + \text{OH}^-$	(10.6)	
Oxygenogenic photosynthesis	Cyanobacteria	$\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{CH}_2\text{O} + \text{O}_2$	(10.7)	Ehrlich (1998)
		$2\text{HCO}_3^- \leftrightarrow \text{CO}_2 + \text{CO}_3^{2-} + \text{H}_2\text{O}$	(10.8)	
		$\text{CO}_3^{2-} + \text{H}_2\text{O} \leftrightarrow \text{HCO}_3^- + \text{OH}^-$	(10.9)	

fresh water is one of the major contributors of the production of carbonates. Cyanobacterial calcification is an extracellular process and occurs on extrapolymeric sheath (EPS) or proteinaceous hydrophobic surface layer (S-layer), which possesses negatively charged sites surrounding the cell (Schultze-Lam and Beveridge, 1994). Generation of alkaline pH at the EPS or S-layer due to carbonic anhydrase (CA) activity promotes the increased CO_3^{2-} concentration at the cell exterior and acts as nucleation site for CaCO_3 precipitation (Jansson and Northen, 2010).

10.2.2.2 Heterotrophic mediated pathway

In heterotrophic mediated pathway, two bacterial carbonate precipitation-enhancing processes are reported (Castanier et al., 1999). First metabolic pathway involves sulfur cycle and second involves nitrogen cycle. First heterotrophic pathway involving sulfur cycle is carried out by SRB via dissimilatory reduction of sulfate. In this process, environment must be anoxic and rich in organic matter, calcium, and sulfate. As the organic matter is degraded by SRB, bicarbonate ions and hydrogen sulfide are produced by bacterial action using SO_4^{2-} as a terminal electron acceptor as shown in Eq. (10.10) (Hammes and Verstraete, 2002; Muyzer and Stams, 2008). Elevation of pH in the surrounding environment due to degasification of hydrogen sulfide in the presence of Ca^{2+} induces the precipitation of calcium carbonate. In natural environment anoxygenogenic sulfide-phototrophic bacteria are present, hydrogen sulfide is used by bacteria, and anaerobically oxidized to sulfur. Uptake of hydrogen sulfide results into pH elevation and favor calcium carbonate precipitation. Precipitation of aragonite crystals on dissolution of gypsum by the action of SRB in Zechstein carbonates (upper Permian) located in central Germany was also reported by Peckmann et al. (1999). It was reported that the replacement of gypsum present in cavities provides the calcium ions for aragonite precipitation and sulfate ions for the metabolic processes of SRB as shown in Eq. (10.11). Degradation of organic matter in anaerobic conditions provides the increased alkaline condition and facilitate the formation of aragonite crystals.



Second heterotrophic pathway in calcium carbonate precipitation involves nitrogen cycle. This pathway is further categorized into three different mechanisms: (1) ammonification of amino acids (presence of organic matter and calcium in aerobic conditions), (2) dissimilatory reduction of nitrate (presence of organic matter, calcium, and nitrate in anaerobic conditions), and (3) urea degradation (presence of organic matter, calcium, and urea in aerobic conditions) (Castanier et al., 2000). In all these three mechanisms, carbonate and bicarbonate ions as well as ammonia (NH_3) are produced as a metabolic end product. Generation of ammonia creates high alkaline pH in the microenvironment of the bacterial cell and decreased H^+ concentration, affecting the carbonate-bicarbonate equilibria shift towards the production of CO_3^{2-} ions.

Presence of calcium ions in the surrounding of bacterial cell results in the precipitation of calcium carbonate. Among the aforementioned mechanisms in heterotrophic pathway, microbially induced calcium carbonate precipitation via urea hydrolysis is widely used in various applications.

10.3 Biocementation in civil engineering

Concrete production is a highly intensive energy-consuming process and presently facing a number of challenges in reducing the carbon footprint and making it more economic. Sustainable technologies in maintaining concrete structures are proving to be a great challenge. Microbially induced carbonate precipitation (MICP) has developed as a novel and sustainable technique in improving the strength and durability of construction materials. The potential and effectiveness of microbial induced biocementation has been used widely in various fields. A number of authors had reported the potential of biocementation property in the application of soil bioclogging (Martinez et al., 2013; Stocks-Fischer et al., 1999), restoration of stone monuments (Le Metayer-Levrel et al., 1999; Jroundi et al., 2014; Micallef et al., 2016), durability enhancement of concrete structures (De Muynck et al., 2008; Achal et al., 2011; Kim et al., 2013; Dhami et al., 2013; Bundur et al., 2015), and crack remediation (Tittelboom et al., 2010; Wiktor et al., 2011; Achal et al., 2013; Da Silva et al., 2015) (see Table 10.2). Stocks-Fischer et al. (1999) reported the urease activity of alkaliphilic bacteria to hydrolyze urea and high pH being the favorable condition for calcite precipitation in porous sand media. Sand column supplemented with live bacterial cells, nutrient broth, and urea- CaCl_2 constituted 30% calcite of total weight of column. Martinez et al. (2013) injected the bacterial cells and subsequent nutrient amendment in half-meter one-dimensional flow sand column. It was reported that change in shear wave velocity from 140 m/s to an average of 600 m/s was observed because of calcium carbonate precipitation resulting into increased shear stiffness of sand. It was suggested that the most important factor to achieve uniform calcium carbonate precipitation is the distribution of bacteria in the column.

Application of MICP technology to prevent the deterioration of monumental stones due to weathering action has been reported by various researchers. Le Metayer-Levrel et al. (1999) reported the reduction in water absorption of limestone samples due to effective biocalcin coating by carbonatogenic bacteria. Tiano et al. (1999) reported the reduction of about 60% in water absorption of limestone samples treated with *Micrococcus* sp. and *Bacillus subtilis* by brushing. Calcite and vaterite crystals precipitated by *Myxococcus xanthus* resulted in efficient protection and consolidation in porous limestone (Rodríguez-Navarro et al., 2003). Jroundi et al. (2014) reported the effective bacterial bioconsolidation produces vaterite (CaCO_3) biocement on the treatment of gypsum plaster collected from the archaeological site “Alcázar de Guadalajara” in Guadalajara, Spain. Reduction in the resorption rate of gypsum material was reported. *Bacillus pumilus* strain isolated from a cave stone effectively precipitated vaterite on the marble (Daskalakis et al., 2015). Micallef et al. (2016) reported the

Table 10.2 Overview of different applications of bacteria in biocementation through biomineralization

Microorganism	Application	Mechanism of treatment	Evaluation	References
<i>Sporosarcina pasteurii</i>	Soil improvement	One-dimensional injection method	Shear wave velocity measurement, CaCO ₃ content, bulk permeability	Martinez et al. (2013)
<i>Bacillus pasteurii</i>	Microbial sand consolidation	Sand slurry column fed by gravity with urea-CaCl ₂ medium	CaCO ₃ content, SEM, XRD	Stocks-Fischer et al. (1999)
<i>Myxococcus xanthus</i>	Archaeological gypsum plasters	Bacterial solution was sprayed till 6 days (twice a day) on the upper surface of sample	Drilling resistance analysis, TGA, XRD, SEM, MIP, TEM, and colorimetric analysis	Jroundi et al. (2014)
<i>Bacillus subtilis</i>	Ancient limestone surface	Brushing or spraying	Drilling resistance analysis, SEM, EDX	Micallef et al. (2016)
<i>Bacillus sphaericus</i>	Mortar and concrete restoration	Specimens were immersed in bacterial culture for 1 day prior to submersion in nutrient medium for 6 days	Compressive strength, sorptivity test, SEM, gas permeability, XRD, chromatic analysis of specimens	De Muynck et al. (2008)
<i>Bacillus</i> sp. CT-5	Cement mortar	Mortar mixture was admixed with NBU medium-bacterial cells and cured with respective medium for 28 days	Compressive strength, water absorption test, and SEM analysis	Achal et al. (2011)
<i>S. pasteurii</i> , <i>B. sphaericus</i>	Concrete specimens	Specimen's top surface was wetted with nutrient medium containing <i>S. pasteurii</i> and <i>B. sphaericus</i> cells separately for 28 days	Water absorption test, SEM-EDX and XRD analysis	Kim et al. (2013)

Continued

Table 10.2 Continued

Microorganism	Application	Mechanism of treatment	Evaluation	References
<i>Bacillus megaterium</i> SS3	Cement mortar blocks and cylinders	Bacterial admixed specimens were cured by spraying with respective medium for 28 days	Water absorption test, SEM-EDX, XRD, MIP, and CaCO ₃ estimation of treated samples	Dhami et al. (2013)
<i>S. pasteurii</i>	Cement mortar	Vegetative cells with UYE medium were added in mortar mixture while casting and cured by submersion in UYE medium with lime till 56 days	Hydration kinetics, compressive strength, TGA, and XRD analysis	Bundur et al. (2015)
<i>B. sphaericus</i>	Crack remediation	Immersed in solution of urea and calcium source for 3 days	Crack width of 0.3 mm and depths of 10.0 and 20.0 mm	Tittelboom et al. (2010)
<i>Bacillus alkalinitrilicus</i>	Crack remediation	Immersed in water for 100 days	Crack width ranging from 0.05 to 1.0 mm	Wiktor et al. (2011)
<i>Bacillus</i> sp. CT-5	Crack remediation	Immersed in urea and CaCl ₂ medium for 28 days	Crack width of 3.0 mm and depths of 13.4, 18.8, and 27.2 mm	Achal et al. (2013)
Nonaxenic ureolytic spores	Crack remediation	Immersed in urea and demineralized water for 4 weeks	Crack healing of width 0.45 mm	Da Silva et al. (2015)

effective biocalcification on *Globigerina* limestone specimen treated with *B. subtilis* strain by spraying and poulticing applications. Uniform bioconsolidation up to 30 mm depth improved surface drilling resistance of limestone, and reduction in water absorption was reported in poultice application.

The potential of MICP technology in cementitious materials to increase the mechanical properties as well as to reduce the permeability properties has been reported by various researchers. Precipitation of CaCO_3 by bacteria inside the cement matrix leads into pore refinement, resulting in reduced permeability and increased strength of concrete structure. Ramachandran et al. (2001) reported enhanced compressive strength of cement mortar cubes on direct incorporation of *Sporosarcina pasteurii* cells inside the cement matrix. De Muynck et al. (2008) reported pronounced decrease in uptake of water and gas permeability in concrete structure because of the use of pure culture of *Bacillus sphaericus* than the use of mixed ureolytic culture. Achal et al. (2011) reported the increase in compressive strength of bacterial treated cement mortar specimen up to 36% in comparison with control specimen. *Bacillus* sp. CT-5 was admixed in the mixture of cement and sand along with nutrient medium and urea- CaCl_2 while casting. Bacterial treated specimen showed six times less water absorption than the control specimen. Kim et al. (2013) investigated the distribution of calcium carbonate precipitation and capillary water absorption of concrete specimens after surface treatment with two bacterial strains, *B. sphaericus* and *S. pasteurii* individually. Denser calcium carbonate crystals and lowest weight increase was reported in specimens treated with *B. sphaericus* strain than the specimens treated with *S. pasteurii* strain. The potential of calcifying bacteria to enhance durability of energy-efficient green building materials was reported by Dhami et al. (2013). Blocks of cement and sand were cured by spraying treatment with NBU medium and bacterial strain *Bacillus megaterium* SS3 for 28 days. Reduction of 40% in water absorption and 31% in the porosity was reported in the biogenic surface-treated specimens in comparison with control specimens. Bundur et al. (2015) reported the increased compressive strength of mortar specimen prepared with incorporation of vegetative bacterial cells than the control specimen. Application of halophilic bacteria *Exiguobacterium mexicanum* isolated from sea water showed 23.5% increase in compressive strength and five times reduction in water absorption on concrete specimens under 5% salt stress condition (Bansal et al., 2016). Kumari et al. (2017) reported 49% increase in compressive strength by using nonureolytic bacteria *Bacillus cohnii*. *Bacillus* sp. CT-5—treated reinforced concrete specimens reduced the rate of corrosion, reduced mass loss, and increased pullout strength than the control specimens (Achal et al., 2012). Kalhori and Bagherpour (2017) reported 30% increase in the compressive strength of bacteria-exposed shotcrete specimens compared to control specimens. Apart from soil bioclogging and concrete restoration, different methodologies have been adopted to heal the concrete cracks with the application of bacteria.

Ramachandran et al. (2001) reported that calcite precipitated during microbial growth enhanced the compressive strength of cracked mortar cubes. The mineralization process was effective in shallow cracks than in deeper ones because of the growth of the bacteria more actively in the presence of oxygen. Tittelboom et al. (2010) investigated the crack healing by immobilizing the bacterial cells of *B. sphaericus* strain

LMG 22257 in silica gel. Standard cracks of width 0.3 mm with two depths of 10 and 20 mm and realistic cracks with width ranging from 0.05 to 0.87 mm were also created in concrete samples. In this experiment, promising results in crack filling and silica-immobilized calcifying bacteria as an efficient self-healing agent were reported. The biochemical healing agent consisting of a mixture of viable but dormant bacteria and organic compounds packed in porous expanded clay particles to heal the cracks was also proposed (Jonkers, 2011).

Wiktor et al. (2011) investigated direct incorporation of bacterial spores and calcium lactate embedded in expanded clay as self-healing agent in concrete. Multiple cracks of widths ranging from 0.05 to 1.0 mm were created in specimen, whereas bacterial based specimen showed complete healing of crack of width 0.42 mm after 100 days of immersion in water. Achal et al. (2013) investigated crack healing process with microbial plugging of artificially created cracks of width 3 mm and different depths of 13.4, 18.8, and 27.2 mm. Successful healing of deepest crack of depth 27.2 mm was reported in microbially treated specimen with bacterial strain *Bacillus* sp. CT-5 along with natural sieved sand. Xu and Yao (2014) investigated nonureolytic bacterially induced CaCO_3 precipitation as a self-healing strategy for concrete cracking by using *B. cohnii* spores. They suggested that incorporation of bacteria and calcium source nutrients as a two-component healing agent in concrete matrix induces CaCO_3 precipitation upon crack formation. Qian et al. (2015) reported the healing of early age cracks in cement-based materials by CA-producing bacteria *Bacillus mucilaginous* L3. Their experimental results showed that the cracks formed at early ages were completely healed (up to 0.4 mm) because of bacterial treatment and the healing effect reduced with the increasing of cracking age. A new biobased powdorous material containing nonaxenic ureolytic spores was developed for self-healing (Da Silva et al., 2015). Cyclic EnRiched Ureolytic Powder (CERUP) produced from a sub-stream of a vegetable treatment plant after drying and grinding contains nonaxenic bacterial culture. Mortar prism admixed with 1% (of cement weight) CERUP was subjected to tensile force to create cracks. CERUP admixed specimen with complete crack closing up to 0.45 mm crack width after 4 weeks was reported.

Field performance of bacteria-based repair system of a two-storey parking garage suffering from cracking and damaged concrete pavement due to freeze/thaw was reported by Wiktor and Jonkers (2015). Cracks that had been treated with the bacteria showed higher freeze/thaw resistance of concrete than the untreated concrete. Tziviloglou et al. (2016) showed various steps taken towards the outdoor applications of biobased self-healing, for the laboratory-scale tests that have shown promising results.

10.4 Biocementation based on CO_2 capture

Biomining of CO_2 by CaCO_3 precipitation is a common phenomenon in marine, freshwater, and terrestrial ecosystems. Photosynthetic microbes, such as cyanobacteria and microalgae, play a prominent role in carbon sequestration by simultaneously capturing CO_2 (Zhu and Dittrich, 2016). Employment of

cyanobacteria in biomineralization offers a novel and self-sustaining strategy in carbon capture and sequestration (CCS). Novel models for point-source CCS based on biomineralization are emerging (Jansson and Northen, 2010). Microalgae have a great potential to be developed as media for biocement production through photosynthetic metabolism (Ariyanti et al., 2012). Some authors had reported the potential of bacteria and microalgae in biocementation of sand stabilization, fugitive dust, and concrete restoration based on CO₂ capture and utilization. Okyay et al. (2016) evaluated the potential of ureolytic consortia in CO₂ sequestration through MICP. In this study, ureolytic consortia were obtained from the “CaveWithout A Name” and “Pamukkale travertines” using 13 different growth media for ureolytic microorganisms. Consortia containing larger abundances of *Sporosarcina*, *Sphingobacterium*, *Stenotrophomonas*, *Acinetobacter*, and *Elizabethkingia* genera were reported for higher CO₂ sequestration and CaCO₃ precipitations. It was suggested that combination of CA enzyme activity with MICP in the consortia might have resulted in a greater rate of CO₂ sequestration. Dhami et al. (2014) also reported the synergistic role of urease (UA) and CA enzyme in biomineralization of calcium carbonate. Calcite precipitation was reported to be significantly reduced when both the enzymes were inhibited in comparison with those of the individual enzyme inhibitions. It was suggested that UA and CA enzymes are important for efficient biomineralization in urea-hydrolyzing bacteria. It was reported that CA plays a role in hydrating CO₂ to bicarbonate, whereas UA activity maintained the alkaline pH promoting calcification process. CO₂ sequestering capability of *B. megaterium* SS3 in biocementation was also reported (Kaur et al., 2016). In this study, urea was replaced with CO₂ flux and under accelerated carbonation in bacterial treated concrete specimen improved compressive strength by 117% as compare with control specimen. Significant reduction in the water absorption was also reported in bacterial treated carbonated specimens. Calcite as well as aragonite crystals were observed in association with the bacterial cells (Fig. 10.1) CO₂ as an alternate was suggested instead of urea in calcium carbonate precipitation paving a way to develop green building structures. Zhan et al. (2016) reported microbial-induced mineralization and cementation of fugitive dust under the action of *Paenibacillus mucilaginosus*. Cementitious material of biological carbonates was prepared by mixing *P. mucilaginosus* bacterial powder and calcium nitrate with water and sprinkled. It was reported that CO₂ absorbed, transformed, and produced carbonate ions under the enzymatic action of *P. mucilaginosus*, resulting in cementation of fugitive dust due to calcite consolidation layer. Zhan and Qian (2017) investigated the stabilization of sand particles using biocementation process based on CO₂ capture and utilization. Bacteria powder of *Paenibacillus* strain and calcium nitrate dissolved in deionized water was sprayed evenly on the surface of sand particles. It was reported that average porosity of samples was reduced from 18.3% to 13.3%. Calcite precipitation due to the enzymatic activity of bacteria and carbonate ion formation due to CO₂ absorption was also reported. Feasibility of application by using cyanobacteria *Synechococcus* PCC8806 in concrete restoration through biomineralization process was reported by Zhu et al. (2015). In this study, concrete cubes with a size of 3 × 3 × 1.5 cm were immersed in 50 mM calcium chloride solution with

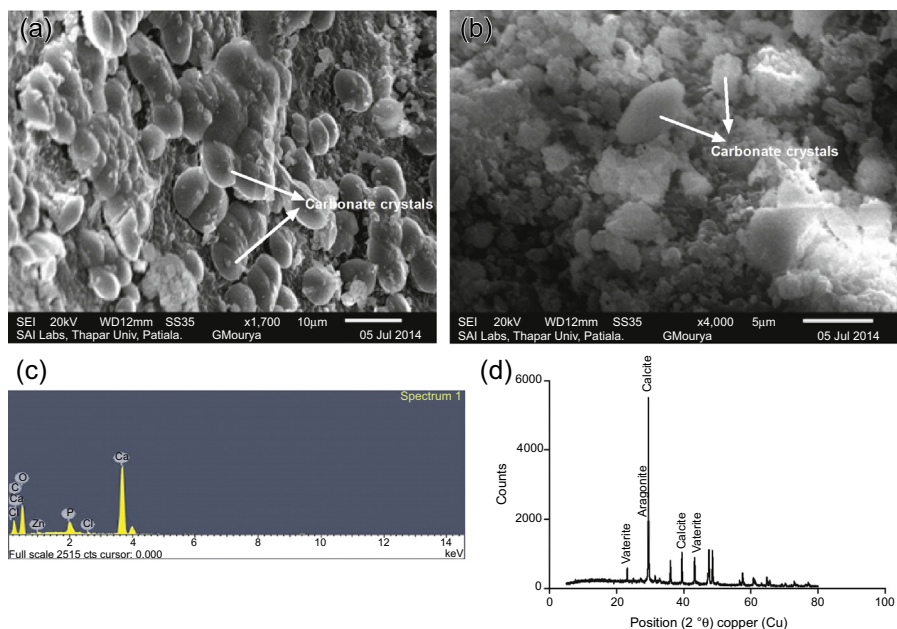


Figure 10.1 Scanning electron microscopic images of carbonate crystals formed by *Bacillus megaterium* through carbonation: (a) carbonate crystals, (b) bacterial cells in association with crystals, (c) energy dispersive X-ray spectrum, (d) X-ray diffraction analysis of carbonate crystals.

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1.5×10^8 cells/mL under shaking condition for 45 days. Thick calcite-cell layer adhering to the treated concrete specimen decreased the water absorption twice as compared with control specimen. [Zhu et al. \(2017\)](#) reported the calcification by cyanobacteria *Gloeocapsa* PCC73106 in mortars. Mortar cubes of a size $50 \times 50 \times 50$ mm were treated abiotically, with live *Gloeocapsa* PCC73106 cells under illumination, with live cells under darkness, and with ultraviolet (UV)-killed cells immersed in the cell solution (with a concentration of 3.5×10^7 cells/mL). Largest amount of precipitates was reported in treatment with live cells under illumination, whereas highest compressive strength, the least water absorption, and the lowest porosity were reported in UV-killed cells. In treatment with live cells, a thin layer of EPS around ([Fig. 10.2\(a\)](#)) cells helped in the attachment to the surface of the mortar ([Fig. 10.2\(b\)](#)). After being exposed to UV light, the UV-killed cells produced more EPS, resulting in larger coverage than live cells. It was reported that UV-killed cells were not metabolically active and EPS enhanced biofilm formation triggering calcium carbonate precipitation on the mortar surface ([Fig. 10.2\(c\)](#) and [\(d\)](#)).

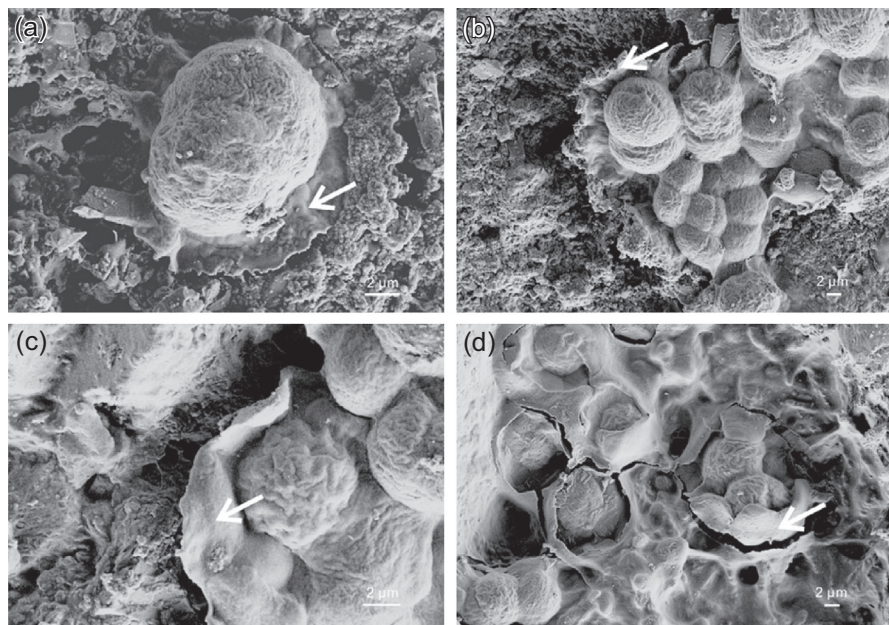


Figure 10.2 Extrapolymeric sheath (EPS) of live and ultraviolet (UV)-killed cells. (a and b) A thin layer of EPS resided between the live cells and the mortar surface. (c and d) A thicker layer of EPS around UV-killed cells.

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10.4.1 Role of microbes in CO₂ sequestration

The ubiquitous presence of microorganisms has the potential to utilize and assimilate CO₂ through different metabolic pathways. Diverse naturally occurring prokaryotic taxa are able to utilize CO₂ through several CO₂ assimilation pathways and convert CO₂ into value-added chemicals or induce calcium carbonate (CaCO₃) precipitation (Hicks et al., 2017). The oceans are huge carbon sinks, and a large proportion is sequestered in the marine sediments in the deep sea mediated by microbial activity. Evidence from the fossil records, association of laminated stromatolites with microbial fossils of photosynthetic cyanobacteria, was reported (Awramik, 1992). Half of the earth photosynthesis is carried out by phytoplankton, which mostly consists of cyanobacteria, and its 25% is accounted by the two marine cyanobacterial genera, *Prochlorococcus* and *Synechococcus*. In CO₂ capturing, photosynthetic microbes, such as cyanobacteria and microalgae, plays a notable role in carbon sequestration (Zhu and Dittrich, 2016). Beachrock formation of El Hamira bay and biogenic calcification due to carbonate precipitation by photosynthetic CO₂ uptake of cyanobacteria was reported by Krumbein in 1979. The potential of halotolerant sulfate-reducing bacteria for carbonate mineralization in geological carbon sequestration during CO₂ injection in subsurface is also reported (Paul et al., 2017).

10.4.2 Enzyme involved in CO₂ sequestration

Carbonic anhydrase (CA) is the first-discovered zinc-containing metalloenzyme that is widespread in animals, plants, and microorganisms that catalyses the conversion of CO₂ and water into bicarbonate (Smith and Ferry, 1999). The CA is widespread in metabolically diverse species of bacteria indicating that this enzyme plays a significant role in concentrating CO₂ (Dhami et al., 2014). In natural process of photosynthetic assimilation of CO₂, CA enzyme acts as biocatalyst (Jansson and Northen, 2010). Potential role of CA enzyme in addressing environmental issues such as reducing carbon emissions through CO₂ sequestration has gained considerable attention. CA enzyme is reported to be a potential tool to sequester CO₂ from emission sources (Bose and Satyanarayana, 2017). It was reported that partially purified CA from *B. pumilus* immobilized on chitosan beads has improved calcium carbonate precipitation than free CA enzyme in carbonation reaction (Wanjari et al., 2011). Immobilization of CA enzyme into alginate beads showed better operational stability by retaining nearly 67% of its initial activity and entrapped CA hydrates CO₂ to bicarbonate and/or carbonate, which on reaction with Ca²⁺ ions, transform into calcite (Yadav et al., 2012). It was reported as a commercialized development for the onsite scrubber for CO₂ sequestration. Zhang et al. (2011) reported the effective absorption of CO₂ into the potassium carbonate solution with biocatalyst CA immobilized into controlled pore glass material. The immobilized enzyme retained at least 60% of their initial activities and significantly improved resistance to concentrations of sulfate (0.4 M), nitrate (0.05 M), and chloride (0.3 M) conditions in flue gas expected in the Integrated Vacuum Carbonate Absorption Process.

10.5 Challenges and Future trends

Biominingalization of CO₂ with the employment of cyanobacteria offers an innovative and self-sustaining strategy in carbon sequestration. Biocementation provides a cost-effective and environment friendly solution to mitigate the increased level of atmospheric CO₂. Calcium carbonates formed after sequestration using bacteria and microalgae will provide industrially valuable and useful by-products that can be implemented in construction material. Microalgae can be easily cultivated and have a great potential in biocement production. As there is rapid increase in the number of patents related to the use of microbes with ureolytic pathway in the construction biotechnology (Dapurkar and Telang, 2017), the critical research needs to be done in making the use of algae as feasible biocementation producer.

Researchers from all over the world have investigated the potential of biomineralization in improving the durability of building materials. However, the qualitative and quantitative evaluation of microbial application of this technology in concrete are still reported in the laboratory scale. Some researchers have also reported encouraging results on the application of bacterial based treatment at field scale. Some limitations that are necessary to be considered for application of this technology at commercial scale have been found. One of the limitations to apply this technology at field scale is use of

laboratory-grade nutrient sources that restrict the use of this technology in several cases. Successful commercialization of the technique requires economical alternatives of the medium ingredients that cost as high as 60% of the total operating costs (Kristiansen, 2001). Use of inexpensive materials as nutrients may help to lower the cost of treatment (Achal et al., 2009, 2011). In field-scale experiments, effective curing method is to be investigated to provide sufficient amount of nutrients and water to the bacterial cells. Further, investigations are needed to improve the microbial technology as an innovative crack healing application for the commercial scale.

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